



HEIMDALL.AI

Seeing what ships hide.

Two cables. Two days. One ship.

Finland–Germany, then Lithuania–Sweden — severed on consecutive days. The same bulk carrier crossed both. The investigation ended with **no charges**. “An accident.”

● Deniability is the weapon

THE STAKES

99%

of the world's data —
finance, the internet, military
command — runs through a
few hundred **cables on the
seabed**. Unguarded.
Unwatched.

THE PATTERN

Nine cuts. Since 2022. One sea.



A **€20,000 anchor** severs a billion-euro artery — and takes weeks to repair. Cheap, deniable, attritional.



We don't lack alerts. We lack prioritization.

The ocean is too big. The sensors that actually see — satellites — are too few. **Where do we point the next one, in the next hour?**

AIS is what a ship **says**.

Every vessel broadcasts who it is. But AIS is **self-reported** — spoof it, fake the identity, or switch it off. The cheapest signal is the one the adversary controls. A ship that wants to cut a cable simply **goes dark**.

SPOOFABLE

FAKEABLE

SWITCH-OFF-ABLE

Three senses. One watch.



MEMORY

Ship Scoring

Every vessel's class, history, sanctions & behaviour — scored for how out-of-character it is now.



EYES

SAR Vision

A fine-tuned detector that finds the ships on the actual water — through cloud, through dark, whatever they broadcast.



THE MAP

Geography

Every cable, pipeline & base as a map of strategic value — what's actually worth protecting.



Knows every ship's story.

Class • track history • sanctions & shadow-fleet •
behavioural coherence. Point-in-time, no look-ahead —
and every score is explainable.

● ...but it only knows what AIS tells it. Lie to it, and it believes the lie.



SENSE 02 • EYES • SENTINEL-1 SAR

Ground truth.
Can't be spoofed.

- ...but a radar blob has no name – and passes are scarce.

THE MODEL THAT MAKES THE EYES WORK

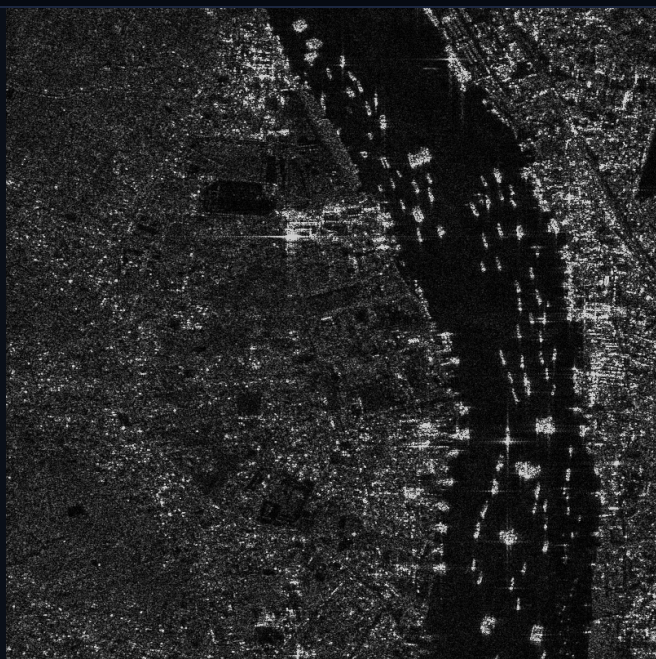
Off-the-shelf AI is blind to radar. Ours isn't.

BASE YOLOV8N · OFF-THE-SHELF

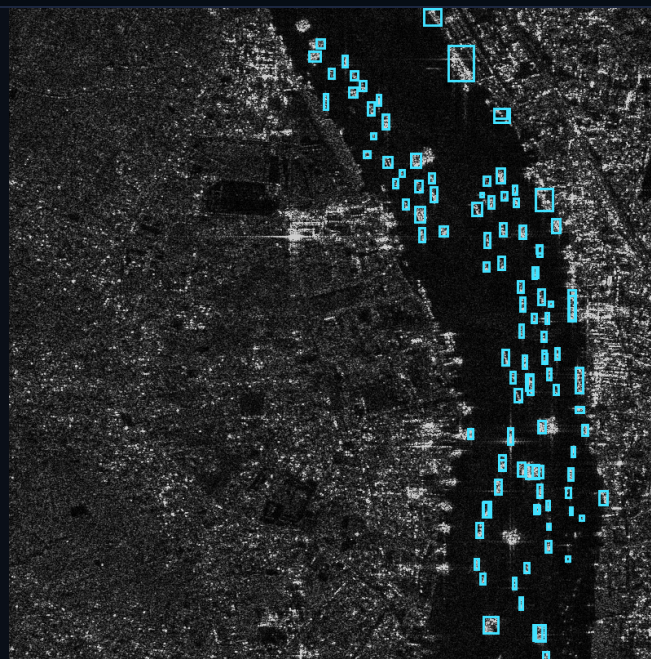
misses the
ships

FINE-TUNED ON HRSID ·
SAR

locks every
hull



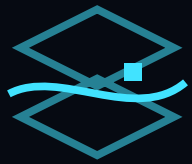
OFF-THE-SHELF · YOLOv8n 0 ships detected



FINE-TUNED · HRSID SAR 105 ships detected

0 → 105

ships found on the same held-out SAR scene, same threshold — off-the-shelf YOLOv8n: 0; our fine-tune: the whole harbour.
~0.91 mAP50 across the HRSID test set.



Knows what's worth protecting.

Turns "a ship slowed down" into "a ship slowed down
right over the C-Lion¹ crossing." 36+ layers: cables,
pipelines, naval bases, chokepoints.

● ...but alone it's just a map – everything near a cable looks busy.

THE TURN

Alone, each one is blind.



Ship Scoring

x LIED TO



SAR Vision

x ANONYMOUS



Geography

x NOISE



AIS is what a ship **says. SAR is what a ship **is**.
The gap between them is the **threat**.**



DEFAULT VIEW

GEOGRAPHY

- ☐ Country borders
- ☐ Territorial seas (12nm)
- ☐ EEZ
- ☐ Rescue zones (SRR)

INFRASTRUCTURE

ENERGY

- ☐ Pipelines
- ☐ Power cables
- ☐ Power plants (incl. nuclear)
- ☐ HVDC converters
- ☐ Terminals / refineries
- ☐ Offshore platforms
- ☐ Wind farms

TELECOM

- ☒ Submarine cables

TRANSPORT

- ☐ Ports (commercial/naval)
- ☐ Shipping lanes (TSS)
- ☐ Anchorages

MILITARY

- ☐ Naval bases
- ☐ Restricted / exercise zones

VESSELS

- ☒ Cargo / bulk
- ☒ Tankers
- ☒ Fishing
- ☒ Passenger / RoPax
- ☒ Military / govt
- ☐ Other
- ☐ Labels

LIVE

C-Lion1 / Yi Peng 3 — replayed on real AIS.

SWITCH TO THE LIVE APP

ALERT FEED

PRIMARY CUE Yi Peng 3 0.83

High vessel risk because SAR/AIS flagged it as a dark approach (no prior track), it ran dark (long AIS gaps) in the lead-up.
SAR +0.50 · behaviour +0.33
09:00:00Z · 55.69N 15.80E · HIGH

Foniks Miljo 1.00

High vessel risk because SAR/AIS flagged an AIS blackout over the incident window, its course was erratic.
behaviour +0.67 · SAR +0.35
09:00:00Z · 55.68N 12.67E · CRITICAL

Tali 1.00

High vessel risk because SAR/AIS flagged it as a dark approach (no prior track), it ran dark (long AIS gaps) in the lead-up.
behaviour +1.00 · SAR +0.50
09:00:00Z · 55.45N 14.61E · CRITICAL

Arklow Abbey 1.00

High vessel risk because it ran dark (long AIS gaps) in the lead-up.
behaviour +1.00
09:00:00Z · 54.16N 12.12E · CRITICAL

SATELLITE TASKING · 3x

C-LION1 / YI PENG 3 · RE-TASK EVERY 3H · CURRENT
11-18 09:00Z · NEXT IN 3H 0M

SAT 1 SAR 0.89

BOX 15.06-15.85E · 55.25-55.70N

Yi Peng 3: SAR dark-approach/blackout, loitering/slow now, anomalous track record

INFERA RISK
LIVE SAR
DARK

VIEW SAR HERE

SAT 2 SAR 0.82

BOX 15.06-15.85E · 55.25-55.70N

No Data: SAR dark-approach/blackout, loitering/slow now, anomalous track record

INFERA RISK
LIVE SAR
DARK

THE VERDICT

Scored blind. Ranked #1.

Point-in-time. No look-ahead. The real investigation took weeks and ended with no charges. Heimdall pulled the culprit out of a thousand ships — the same morning.

SAT 1

SAR

0.89

BOX 15.06-15.85E · 55.25-55.70N ·
SATELLITE TASKING · 2024-11-18 09:00Z

Yi Peng 3 — SAR dark-approach,
loitering over the cable now,
anomalous track record.

INFRA



RISK



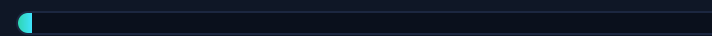
LIVE



SAR



DARK



WHY NOW

The doctrine exists. The cueing layer doesn't.

NATO

**Critical Undersea
Infrastructure Cell**

stood up 2024

FRANCE

**Seabed-warfare
doctrine**

maîtrise des fonds marins

EU

**Cable-security
mandate**

recommendation 2024

Everyone tracks ships. No one tasks the sensor — and the sensor is the scarce thing. That tasking decision is the layer we own. Open data → we deploy day one.



HEIMDALL.AI

AIS is what a ship **says**. SAR is what a ship **is**.
The gap is the **threat**.

Looking for one design partner — a navy or a cable operator — to point Heimdall at live water.